

Streamlit Complete UI Functions Guide

This guide covers important Streamlit UI functions with syntax, purpose, usage, and examples.

st.title()

Why Used: Main page title

Syntax: `st.title('Student Dashboard')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.header()

Why Used: Section heading

Syntax: `st.header('Data Analysis')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.subheader()

Why Used: Subsection heading

Syntax: `st.subheader('Sales Report')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.text()

Why Used: Display plain text

Syntax: `st.text('Hello World')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.write()

Why Used: Most versatile output function

Syntax: `st.write(df)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.markdown()

Why Used: Display markdown formatting

Syntax: `st.markdown('**Bold Text**')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.code()

Why Used: Display code block

Syntax: `st.code('print(Hello)')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.metric()

Why Used: Show KPI metrics

Syntax: `st.metric('Revenue', '■50K', '+10%')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.dataframe()

Why Used: Interactive dataframe

Syntax: `st.dataframe(df)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.table()

Why Used: Static table

Syntax: `st.table(df.head())`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.image()

Why Used: Display images

Syntax: `st.image('img.png')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.audio()

Why Used: Play audio

Syntax: `st.audio('song.mp3')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.video()

Why Used: Play videos

Syntax: `st.video('video.mp4')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.button()

Why Used: Create button

Syntax: `st.button('Submit')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.checkbox()

Why Used: Create checkbox

Syntax: `st.checkbox('Accept Terms')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.radio()

Why Used: Single selection

Syntax: `st.radio('Gender',['Male','Female'])`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.selectbox()

Why Used: Dropdown selection

Syntax: `st.selectbox('City',cities)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.multiselect()

Why Used: Multiple selections

Syntax: `st.multiselect('Skills',skills)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.slider()

Why Used: Range/value slider

Syntax: `st.slider('Age',18,60)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.text_input()

Why Used: Text input box

Syntax: `st.text_input('Name')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.text_area()

Why Used: Multi-line input

Syntax: `st.text_area('Feedback')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.number_input()

Why Used: Numeric input

Syntax: `st.number_input('Salary')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.date_input()

Why Used: Date picker

Syntax: `st.date_input('DOB')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.file_uploader()

Why Used: Upload files

Syntax: `st.file_uploader('Upload CSV')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.sidebar

Why Used: Sidebar UI

Syntax: `st.sidebar.selectbox(...)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.columns()

Why Used: Create columns

Syntax: `col1,col2=st.columns(2)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.tabs()

Why Used: Create tabs

Syntax: `tab1,tab2=st.tabs(['EDA','ML'])`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.expander()

Why Used: Collapsible section

Syntax: with `st.expander('Details')`:

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.progress()

Why Used: Progress bar

Syntax: `st.progress(50)`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.spinner()

Why Used: Loading spinner

Syntax: with `st.spinner('Loading')`:

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.success()

Why Used: Success message

Syntax: `st.success('Done')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.warning()

Why Used: Warning message

Syntax: `st.warning('Check Input')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.error()

Why Used: Error message

Syntax: `st.error('Invalid Data')`

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.info()

Why Used: Information message

Syntax: st.info('Dataset Loaded')

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.plotly_chart()

Why Used: Display Plotly chart

Syntax: st.plotly_chart(fig)

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.pyplot()

Why Used: Display Matplotlib chart

Syntax: st.pyplot(fig)

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

st.download_button()

Why Used: Download files

Syntax: st.download_button(...)

Used frequently while building professional Streamlit dashboards, ML apps, NLP apps, and data science projects.

Typical Streamlit App Structure

1. Import libraries
2. Create title/header
3. Take user input
4. Load data/model
5. Display charts/tables
6. Show prediction/results
7. Allow download/export